Mock test

Q1

a) Here, $m = \frac{1}{r_{CD}} = \frac{I_{D_2} - I_{D_1}}{V_{D_2} - V_{D_1}}$

$I_{D_2} = 1.409 \times 10^{-3} A$
$I_{D_1} = 466.006 \times 10^{-6} A$
$V_{D_2} = 2V$
$V_{D_1} = 1V$

$$= \frac{1.409 \times 10^{-3} - 466.006 \times 10^{-6}}{2 - 1}$$

$$\Rightarrow \frac{1}{r_{CD}} = 9.429 \times 10^{-4}$$

$$\Rightarrow r_{CD} = 1060.45\ \Omega$$

b) $V_T = \frac{KT}{q} = \frac{1.38 \times 10^{-23} \times 300}{1.6 \times 10^{-19}} = 0.0259V$

c) Taking two lower points,

$I_{D_1} = 22.264 \times 10^{-6} A$ $I_{D_2} = 131.231 \times 10^{-6} A$
$V_{D_1} = 400 \times 10^{-3} V$ $V_{D_2} = 600 \times 10^{-3} V$

We know that,

$I_{D_1} = I_S e^{\frac{V_{D_1}}{nV_T}}$, $I_{D_2} = I_S e^{\frac{V_{D_2}}{nV_T}}$

$$\frac{I_{D_2}}{I_{D_1}} = e^{\frac{V_{D_2} - V_{D_1}}{nV_T}}$$

$$\Rightarrow \frac{V_{D_2} - V_{D_1}}{nV_T} = \ln\left(\frac{I_{D_2}}{I_{D_1}}\right)$$

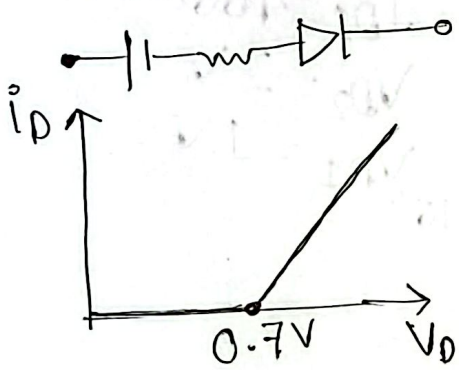
$$\Rightarrow n = \frac{V_{D_2} - V_{D_1}}{V_T \ln\left(\frac{I_{D_2}}{I_{D_1}}\right)} = \frac{600 \times 10^{-3} - 400 \times 10^{-3}}{0.0259 \times \ln\left(\frac{131.231}{22.264}\right)}$$

$$= 4.35,$$

$$\therefore I_S = I_{D_1} e^{-\frac{V_{D_1}}{nV_T}} = 22.264 \times 10^{-6} \times e^{-\frac{400 \times 10^{-3}}{4.35 \times 0.0259}}$$

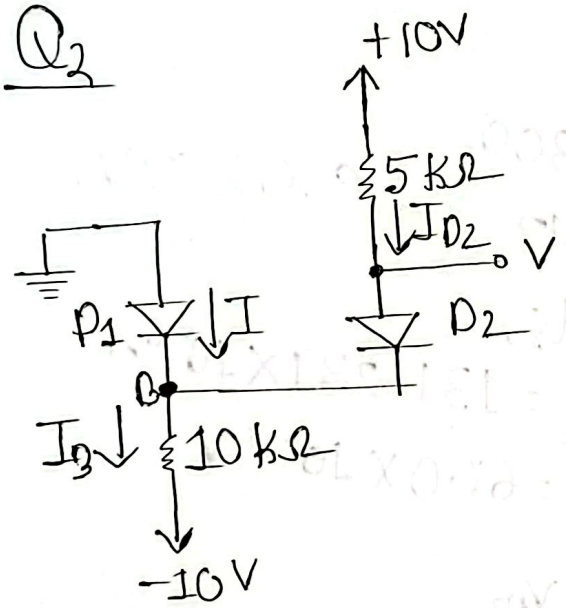
$$= 6.39 \times 10^{-7} A,$$

e) PWL (Piece-Wise Linearc model)



A fixed drop voltage with a small resistorc. More accurcate than CVD and Ideal diode model.

Q2



Soln:

Assuming both diodes are conducting,

$$I_{D_2} = \frac{10-0}{5K} = 2\,mA$$

$$I_3 = \frac{0-(-10)}{10K} = 1\,mA$$

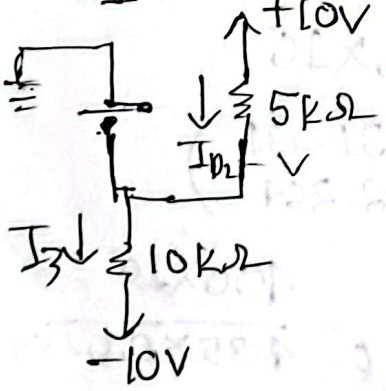
Applying KCL at node-B,

$$I + I_{D_2} = I_3$$

$$\Rightarrow I = I_3 - I_{D2} = (1-2)\,mA = -1\,mA.$$

Diode current can't be negative. Ourc assumption is wrong.

Now, assuming D_1 is off and D_2 is ON.



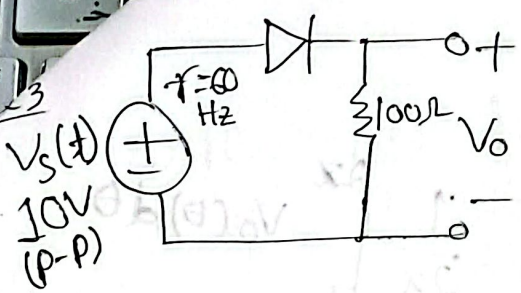
$$I_3 = I_{D_2} = \frac{10-(-10)}{15K}$$

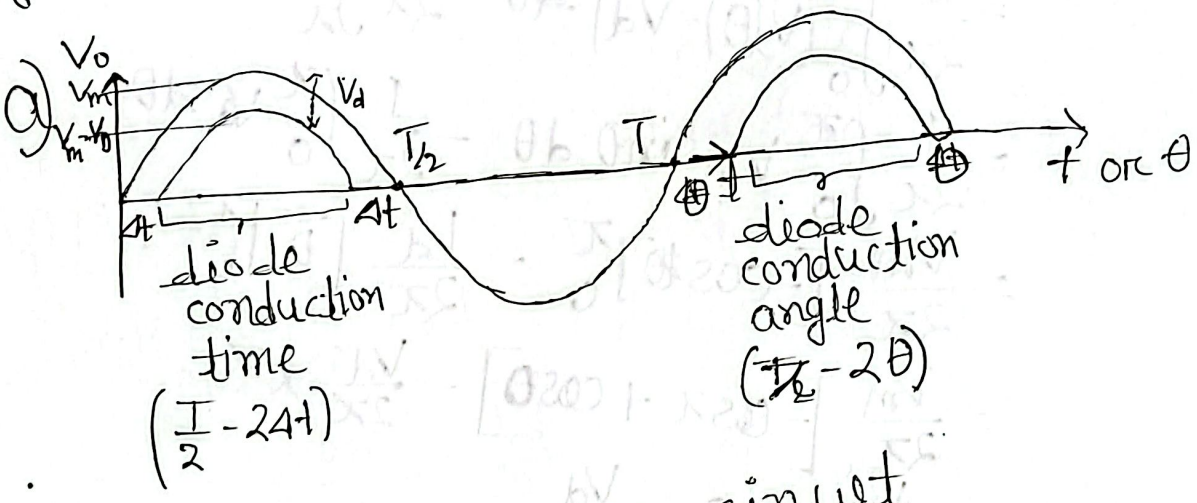
$$= 1.33\,mA$$

$I_{D1} = 0$ $\therefore$ D_1 is OFF.

$$V = 10 - (5 \times 1.33)$$

$$= 3.35\,V.$$





Here,

input: $V_s(t) = V_m \sin \omega t$

$V_s(\theta) = V_m \sin \theta$

Output: $V_o(t) = 0$ if $V_s < V_d$

$V_o(t) = V_s(t) - V_d$ if $V_s \geq V_d$

$V_o(\theta) = V_s(\theta) - V_d$ if $V_s \geq V_d$

∴ Avercage value of the output voltage,

$$V_{o(avg)} = \frac{1}{T}\int_0^T V_o(t)\,dt$$

$$= \frac{1}{T}\int_0^{T/2} V_o(t)\,dt + \frac{1}{T}\int_{T/2}^{T} V_o(t)\,dt$$

$$= \frac{1}{T}\int_0^{T/2} V_s(t)\,dt - \frac{1}{T}\int_0^{T/2} V_d\,dt$$

$$= \frac{1}{T}\int_0^{T/2} V_m \sin \omega t\,dt - \frac{V_d}{T}\int_0^{T/2} dt$$

$$= \frac{V_m}{T\omega}\left[-\cos \omega t\right]_0^{T/2} - \frac{V_d}{T}\left[t\right]_0^{T/2}$$

$$= \frac{V_m \times 1}{T \times 2\pi}\left[-\cos\frac{2\pi}{T}\times\frac{T}{2} + \cos 0\right] - \frac{V_d}{T}\times\frac{T}{2}$$

$$= \frac{V_m}{2\pi}\times 2 - \frac{V_d}{2} = \frac{V_m}{\pi} - \frac{V_d}{2}\ \text{(Ans)}$$

again,

$$V_{o(avg)} = \frac{1}{2\pi}\int_0^{2\pi} V_o(\theta)\,d\theta$$

$$= \frac{1}{2\pi}\int_0^{\pi}(V_s(\theta) - V_d)\,d\theta - \frac{1}{2\pi}\int_{\pi}^{2\pi} V_o(\theta)\,d\theta$$

$$= \frac{1}{2\pi}\int_0^{\pi} V_m \sin\theta\,d\theta - \frac{1}{2\pi}\int_0^{\pi} V_d\,d\theta$$

$$= \frac{V_m}{2\pi}\left[-\cos\theta\right]_0^{\pi} - \frac{V_d}{2\pi}\left[\theta\right]_0^{\pi}$$

$$= \frac{V_m}{2\pi}\left[-\cos\pi + \cos\theta\right] - \frac{V_d}{2\pi} \times \pi$$

$$= \frac{V_m}{2\pi} \times 2 - \frac{V_d}{2}$$

$$= \frac{V_m}{\pi} - \frac{V_d}{2},$$

d) Diode conduction time, $= \frac{T}{2} - 2\Delta t$

We know that,

$$V_m \sin\omega\Delta t = V_d$$

$$\Rightarrow \omega\Delta t = \sin^{-1}\left(\frac{V_d}{V_m}\right)$$

$$\Rightarrow \omega\Delta t = 0.1404 \text{ rad}$$

$$\Rightarrow \Delta t = \frac{0.1404}{2\pi \times 60} = 3.72 \times 10^{-4} \text{ s},$$

Diode conduction time, $= \frac{T}{2} - 2\Delta t$

$$= \frac{1}{2\times 60} - 2 \times 3.72 \times 10^{-4}$$

$$= 7.58 \text{ ms},$$

e) $V_m \sin\theta = V_d$

$$\Rightarrow \theta = \sin^{-1}\left(\frac{0.7}{5}\right) = 8.01^\circ$$

$$= 0.1404 \text{ rad}$$

$\therefore$ Diode conduction angle $= (\pi - 2\theta)$

$$= \pi - 2 \times 0.1404$$

$$= 2.86 \text{ rad} = 163.90^\circ,$$

g) Diode conduction percentage $= \frac{\frac{T}{2} - 2\Delta t}{T} \times 100\%$

$= \frac{7.58 \times 10^{-3}}{0.01667} \times 100\%$

$= 45.48\%$

h) PIV $= V_m = 5\ V$

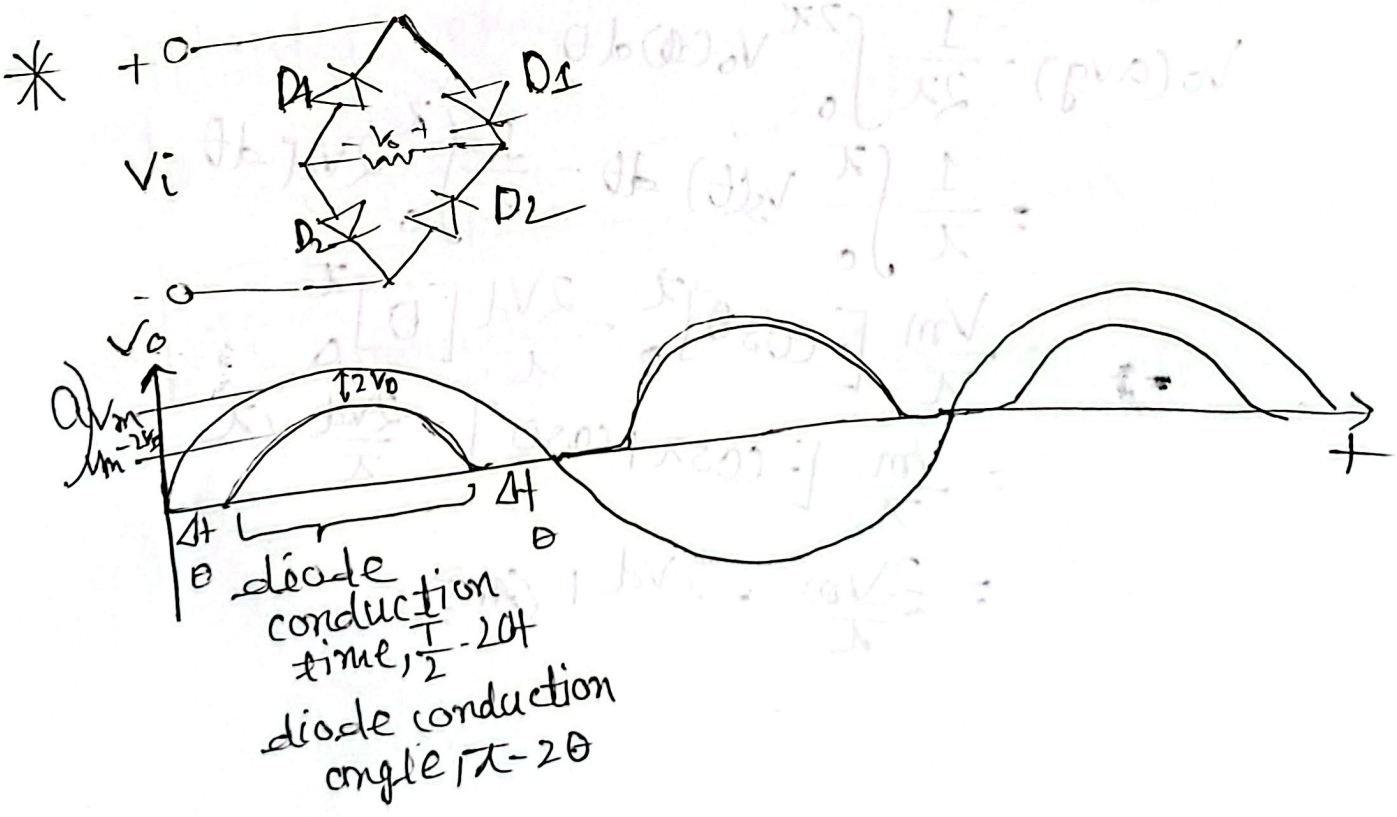
i) Peak diode current, $I_L = \frac{V_m - V_d}{R} = \frac{5 - 0.7}{100} = 43\ mA$

j) $V_{o(avg)} = \frac{V_m}{\pi} - \frac{V_d}{2}$

$= \frac{5}{\pi} - \frac{0.7}{2}$

$= 1.24\ V$

$\pi - 2\theta = \omega\left(\frac{T}{2} - 2\Delta t\right)$



c, d, e) same

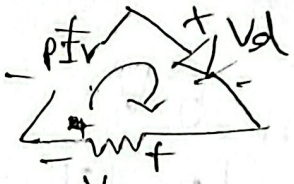
f) Full cycle diode conduction time $= 2\left(\frac{T}{2} - 2\Delta t\right)$

g) Full cycle diode conduction angle $= 2(\pi - 2\theta)$

i) $PIV = V_m - V_D$

j) Peak diode current $= \frac{V_m - 2V_D}{R_f}$

k) $V_{o(avg)} = \frac{2V_m}{2\pi} - 2V_D$



$PIV - V_d - V_o = 0$

$\Rightarrow PIV = V_d + V_m - 2V_d$

$\boxed{\Rightarrow PIV = V_m - V_d}$

* $V_s(\theta) = V_m \sin\theta$

$V_o(\theta) = 0$; $V_s < V_D$

$V_o(\theta) = V_s(\theta) - 2V_D$; $V_s \geq V_D$

$$V_{o(avg)} = \frac{1}{2\pi}\int_0^{2\pi} V_o(\theta)\, d\theta$$

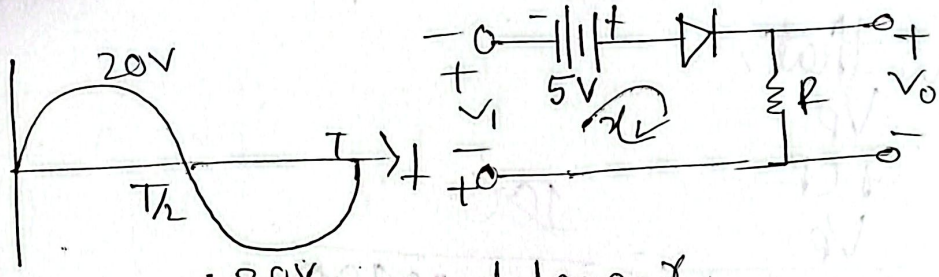
$$= \frac{1}{\pi}\int_0^{\pi} V_s(\theta)\, d\theta - \frac{1}{\pi}\int_0^{\pi} 2V_d\, d\theta$$

$$= \frac{V_m}{\pi}\left[-\cos\theta\right]_0^{\pi} - \frac{2V_d}{\pi}\left[\theta\right]_0^{\pi}$$

$$= \frac{V_m}{\pi}\left[-\cos\pi + \cos 0^\circ\right] - \frac{2V_d}{\pi} \times \pi$$

$$= \frac{2V_m}{\pi} - 2V_d \quad \text{(Ans)}$$

Q4



Applying KVL at loop-2,

$$V_i + 5 - V_o = 0$$

$$\Rightarrow V_i = -5V$$ → Transition voltage

again,

$$V_i + 5 - V_o = 0$$

$$\Rightarrow V_o = 20 + 5 = 25V$$ (+) cycle

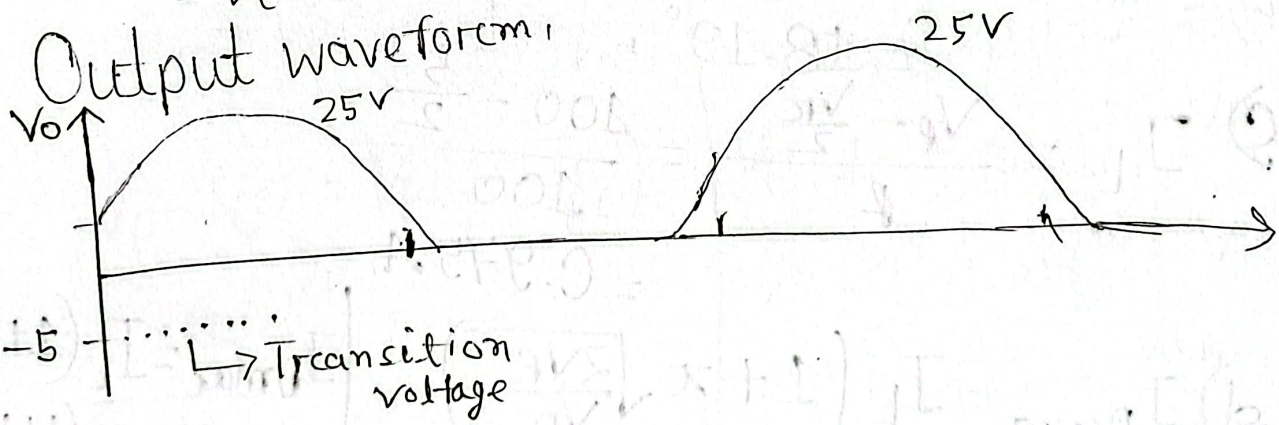
at (-) cycle,

$V_o = 0$ V

~~$V_o = 0$~~

~~$-V_i + 5$~~

Output waveform:



a) We know that,

$$V_{rc} = \frac{V_p}{fCR}$$

$$\Rightarrow C = \frac{V_p}{fV_{rc}R} = \frac{100}{60 \times 5 \times 100}$$

$$= 3.33 \times 10^{-3} F$$

b) Conduction angle,

$$\omega \Delta t = \theta$$

$$V_m \cos(\omega \Delta t) = V_p - V_{rc}$$

$$\Rightarrow \omega \Delta t = \cos^{-1}\left(\frac{V_p - V_{rc}}{V_p}\right)$$

$$= \cos^{-1}\left(\frac{95}{100}\right)$$

$$= 0.317 \text{ rad}$$

$$= 18.19^\circ,$$

c) $$I_L = \frac{V_p - \frac{V_{rc}}{2}}{R} = \frac{100 - \frac{5}{2}}{100}$$

$$= 0.975 A$$

d) $$I_{Davg} = I_L\left(1 + \pi\sqrt{\frac{2V_p}{V_{rc}}}\right)$$

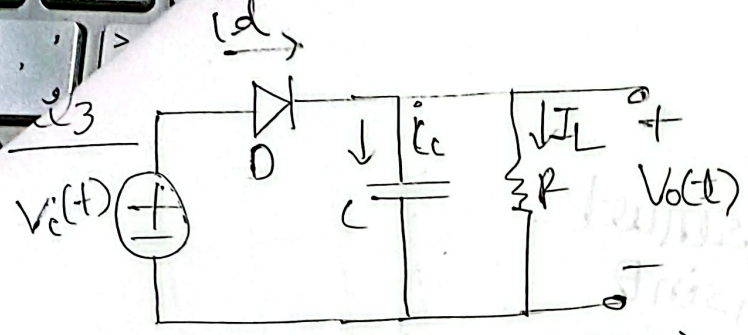
$$= 0.975\left(1 + \pi\sqrt{\frac{2 \times 100}{5}}\right)$$

$$= 20.34 A.$$

$$I_{Dmax} = I_L\left(1 + 2\pi\sqrt{\frac{2V_p}{V_{rc}}}\right)$$

$$= 0.975 \times \left(1 + 2\pi\sqrt{\frac{2 \times 100}{5}}\right)$$

$$= 39.71 A,$$



$V_D = 0.7V$, $V_i = 70.7$ (RMS) $= 99.98$ V, $V_{rc} = 5$ V, $f = 60$ Hz, $R = 100\Omega$, $T = 0.016675$

*Derive,

$$V_{rc} = \frac{V_p}{fCR}$$

$V_o = V_p e^{-t/RC}$ [during discharging]

$\Rightarrow V_p - V_{rc} \cong V_p e^{-T/RC}$ [at the end of discharging]

for small $RC >> T$,

$$e^{-T/RC} = 1 - \frac{T}{RC}$$

$$\therefore V_p - V_{rc} = V_p\left(1 - \frac{T}{RC}\right)$$

$$\Rightarrow V_{rc} = \frac{V_p T}{RC}$$

$$\boxed{\Rightarrow V_{rc} = \frac{V_p}{fCR}}$$

$V_p \cos(\omega\Delta t) = V_p - V_{rc}$

For small angle,

$\cos(\omega\Delta t) = 1 - \frac{1}{2}(\omega\Delta t)^2$

$V_p\left(1 - \frac{1}{2}(\omega\Delta t)^2\right) = V_p - V_{rc}$

$\Rightarrow \frac{V_p}{2}(\omega\Delta t)^2 = V_{rc}$

$\Rightarrow (\omega\Delta t)^2 = \frac{2V_{rc}}{V_p}$

$\Rightarrow \omega\Delta t = \sqrt{\frac{2V_{rc}}{V_p}}$

* Full wave Bridge rectifiers,

Input, $V_s(t) = V_m \sin \omega t$

$V_s(\theta) = V_m \sin \theta$

Output,

$V_o(t) = 0 \; ; V_s < V_D$

$V_o(t) = V_s(t) - 2V_D \; ; V_s \geq V_D$

$V_o(\theta) = V_s(\theta) - 2V_D \; ; V_s \geq V_D$

Average value of output voltage,

$$V_{o(avg)} = \frac{1}{T}\int_0^T V_o(t)\,dt$$

$$= \frac{1}{T/2}\int_0^{T/2} (V_s(t) - V_D)\,dt$$

$$= \frac{2}{T}\int_0^{T/2} V_m \sin \omega t - \frac{2}{T}\int_0^{T/2} 2V_D\,dt$$

$$= \frac{2V_m}{T\omega}\left[-\cos \omega t\right]_0^{T/2} - \frac{4V_D}{T}\left[t\right]_0^{T/2}$$

$$= \frac{2V_m \times T}{T \times 2\pi}\left[-\cos\frac{2\pi}{T} \times \frac{T}{2} + \cos 0^\circ\right] - \frac{4V_D}{T} \times \frac{T}{2}$$

$$= \frac{V_m}{\pi} \times 2 - 2V_D$$

$$= \frac{2V_m}{\pi} - 2V_D \text{ (Ans)}$$